a GPCR–arrestin structures by engagement mode

All 36 structures in Table 1 of the paper (He et al. 2026). Mode: how arrestin sits on the receptor.
Core is 29 of 36 as classed in Table 1; four receptors carry a dagger. Each other mode has one receptor.

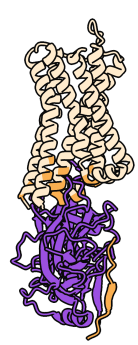
Colour Arr1 = arrestin-1 (rhodopsin, X-ray) · βarr1 = β-arrestin 1 (arrestin-2) · βarr2 = β-arrestin 2 (arrestin-3)
Shape cryo-EM · X-ray (XFEL = X-ray free-electron laser) · hollow: 5DGY, XFEL in the paper, synchrotron in its PDB record
Size Size repeats the resolution printed under each ID: bigger = sharper.
Rows: non-Core receptors first, in column order, then in the paper's table order. Blank cell: no entry. Ring: drawn in panel b.

Receptor	Core 29 entries, 13 receptors	Tail 2 entries	Loop-engaged 2 entries	Side-engaged 2 entries	Membrane-anchored 1 entry
GCGR		**8JRV** 3.30 Å · **8JRU** 3.50 Å			
NTSR1	**6PWC** 4.90 Å · **6UP7** 4.20 Å · **8ZYT** 2.65 Å · **9M0D** 3.41 Å		**8ZYU** 2.65 Å · **8ZYY** 2.83 Å		
GPR52				**9IJR** 3.86 Å · **9IJS** 3.64 Å	
ACKR3					**9E82** 3.40 Å
Rhodopsin	**4ZWJ** 3.30 Å · **5DGY** 7.70 Å · **5W0P** 3.01 Å				
M2R	**6U1N** 4.00 Å				
Turkey β_1AR	**6TKO** 3.30 Å				
V2R	**7R0C** 4.73 Å				
5-HT$_{2B}$R	**7SRS** 3.30 Å				
CB1R	**8WRZ** 3.60 Å · **8WU1** 3.20 Å				
mGlu3R†	**9II2** 3.70 Å · **9II3** 3.90 Å				
APJR†	**9KUV** 3.21 Å · **9KUW** 3.49 Å · **9KUX** 3.57 Å				
mGlu8R†	**9MBA** 3.14 Å				
GPR1	**9UYH** 3.30 Å · **9UYI** 3.20 Å · **9UYJ** 3.30 Å · **9UYL** 3.30 Å **9UYN** 2.90 Å · **9UYM** 3.50 Å				
MOR	**9WSV** 2.80 Å · **9WSX** 2.80 Å				
PTH1R†	**9LXR** 3.07 Å · **9LXP** 3.20 Å				

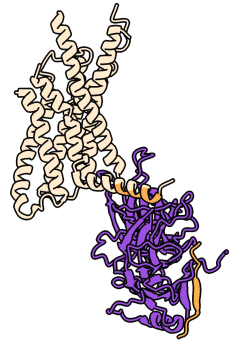
† Table 1 says Core; the text describes them differently.
PTH1R: tail- and core-engaged (elsewhere both core).
mGlu3R, mGlu8R: shallow, peripheral interface.
APJR dimers: tail-dominant or shallow-core.

b Three entries in one view

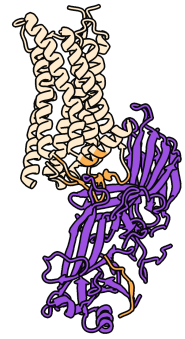
receptor · arrestin (βarr1) · receptor residues within 4.5 Å of arrestin



Core, 9WSV
MOR + V2R tail, βarr1
32 orange residues
12 in the V2R tail



Tail, 8JRV
GCGR + V2R tail, βarr1
21 orange residues
12 in the V2R tail



Loop-engaged, 8ZYU
NTSR1, βarr1
30 orange residues

Core: the arrestin finger loop inserts deep into the receptor's transmembrane core.
Tail: held mainly by the receptor's phosphorylated C-terminus; core stays open.
Loop-engaged: receptor loop ICL3 inserts into arrestin; finger loop stays outside the core.
Side-engaged: arrestin binds sideways via ICL1 and helix 8; no core insertion.
Membrane-anchored: held by the C-terminus; finger loop enters the lipid or micelle boundary.
Definitions are shortened from the paper. ICL: intracellular loop.
Orange: loops, helix ends, helix 8, C-terminal tail. Loops are not labelled.
One receptor, one arrestin per render. 9WSV, 8JRV: the receptor ends in the V2R tail.

Marks follow Table 1 as published; panel b is drawn from PDB coordinates.

After Fig. 2 and Table 1 of He et al., Cell Discovery 2026, doi:10.1038/s41421-026-00909-1, CC BY 4.0. Data: Table 1, PDB. Redesigned; values as published.